

Carrefour Taiwan: Optimizing customer engagement and online shopping experiences through cloud transformation

Tell us your challenge. We're here to help.

[Contact us \(/contact\)](#)

ABOUT CARREFOUR



About Carrefour Taiwan

As a part of its digital transformation effort, Carrefour Taiwan leveraged Google Cloud to implement AI-powered precision marketing, offer r

In 1989, world-leading retail brand Carrefour established its first store in Taiwan. As of June 2022, Carrefour Taiwan has more than 100 supermarkets and retail stores across the country. Looking to build a comprehensive retail ecosystem, Carrefour Taiwan embarked on their digital transformation journey in 2015, and started offering more diverse shopping services with an ecommerce platform and a mobile app, which currently has more than five million downloads.

reliable online retail services and optimized internal systems work efficiency.

Industries: Retail & Consumer Goods

Location: Taiwan

Google Cloud results

- Reaches 2.64X higher return on ad spend (ROAS) through precision marketing supported by AutoML
- Supports 6X faster system operating speed with a local data center
- Helps save 40% of operational costs with more cost-effective

Lower cost
action
advertising
40%

Kubernetes Engine

Google Cloud (<https://cloud.google.com/>)

Cloud AutoML (<https://cloud.google.com/automl/>)

BigQuery (<https://cloud.google.com/bigquery/>)

Cloud Run (<https://cloud.google.com/run/>)

Cloud SQL (<https://cloud.google.com/sql/>)

Kubernetes Engine

(<https://cloud.google.com/kubernetes-engine/>)

SAP on Google Cloud

(<https://cloud.google.com/solutions/sap/>)

cloud
infrastructure

- Eliminates
frequent
downtime
caused by
high traffic
with easily
scalable
computing
resources

As online shopping continues to grow and evolve, people's consumption habits have significantly changed over the past decade. To meet consumers' new needs and better compete with digital merchants, established retail brands are transforming the way they operate, and expanding their services with the help of the latest technologies like cloud and artificial intelligence (AI).

Carrefour Taiwan (<https://www.carrefour.com.tw/>) is in the process of modernizing its operations and services through digital transformation, which it embarked on in 2015. Since its parent company Carrefour, a world-leading (<https://nrf.com/blog/look-2022-top-50-global-retailers>) retail brand headquartered in France, entered the Taiwanese market in 1989, Carrefour Taiwan has expanded its presence in the country to more than 100 supermarkets and retail stores. Aiming at building a retail ecosystem across physical and virtual channels, the company has also launched an

ecommerce platform and a mobile app to offer more diverse shopping experiences. As of June 2022, its app had more than five million downloads.

"Our goal is to provide services to our customers no matter where they are and at their greatest convenience. To achieve that, full digital transformation is necessary, and we're always looking for ways to make the best of new technologies in the transformation process," notes Henry Ting, Digital Chief Technology Officer at Carrefour Taiwan.

As a part of its digital transformation effort, Carrefour Taiwan set a goal in 2018 to reach exponential growth in its ecommerce revenues. As online advertising plays a big role in driving up ecommerce sales, the retailer needed to implement more cost-effective advertising strategies and decided to use AI technology to build machine learning (ML) models for precision marketing.

At the same time, Carrefour Taiwan wanted to move its on-prem internal systems to cloud and enhance the performance of its ecommerce platform and mobile app that were hosted in a public cloud platform's data center in Hong Kong, because the network connection was not ideal. After comparing different offerings, Carrefour Taiwan chose Google Cloud (<https://cloud.google.com/>) as its only cloud provider, based on its high integration between its ML products and Google Ads (https://ads.google.com/intl/en_us/home/#!/), which the retailer highly relies on for online advertising. Google

Cloud's data center in Taiwan also means that it offers faster and more stable connectivity.

"Our cloud transformation plan involves various aspects, including ML model training and IT infrastructure modernization," explains Ting. "Google Cloud perfectly meets all our needs with its high-performance cloud tools and its support of online advertising campaign execution. We can fully count on Google Cloud to run all our businesses in the cloud."

"Our goal is to
provide services
to our
customers no
matter where
they are and at
their greatest
convenience. To
achieve that, full
digital
transformation
is necessary,
and we're always
looking for ways
to make the best

of new
technologies in
the
transformation
process."

—Henry Ting, Digital Chief
Technology Officer,
Carrefour Taiwan

Enhancing customer engagement with precision marketing supported by AutoML

As the first step of its adoption of Google Cloud, Carrefour Taiwan uploaded its customer data to AutoML (<https://cloud.google.com/automl>) in 2019, and built four ML models for customer lifetime value prediction, segmentation, conversion rate prediction and churn rate prediction. At that time, the retailer didn't have professionals who were experienced in setting up complex ML models. However, Google Cloud Professional Service Organization helped Carrefour Taiwan map out the development steps of the ML models, and the Google Ads team offered advice on defining the key performance indicators for advertising results. This collaboration enabled Carrefour Taiwan's ML models to deliver useful insights for the company's marketing campaigns.

For example, based on the analytics results generated by the customer lifetime value prediction

model, Carrefour Taiwan is able to identify its high-value customers. By delivering ads primarily to these customers, the company has lowered its cost per action of online advertising by 40%. Meanwhile, the customer segmentation model provides data that enables Carrefour Taiwan to understand its customers' preferences better and send out more personalized ads, leading to higher returns on its ad investments.

"With our ML models powered by AutoML, we know more about our customers' consumption habits and preferences, and are therefore able to run more cost-effective advertising campaigns while reaching better customer engagement," says Ting. "Overall, our return on ad spend is 2.64 times higher compared to before."

"With our ML
models powered
by AutoML, we
know more
about our
customers'
consumption
habits and
preferences, and
are therefore

able to run more
cost-effective
advertising
campaigns while
reaching better
customer
engagement.

Overall, our
return on ad
spend is 2.64
times higher
compared to
before."

—Henry Ting, Digital Chief
Technology Officer,
Carrefour Taiwan

Achieving faster system operating speeds and connection stability with a local data center

After building its ML models on AutoML, Carrefour Taiwan started moving its internal systems from an on-prem environment in France, where Carrefour's headquarters is situated, and its ecommerce platform and mobile app from another public cloud platform, to Google Cloud. The retailer completed the migration in July 2021, and now deploys its enterprise resource planning, human resource and

supply chain management system on SAP on Google Cloud, while running its ecommerce platform with containers on Cloud Run (<https://cloud.google.com/run>) and its mobile application with servers on Google Kubernetes Engine (<https://cloud.google.com/kubernetes-engine>) (GKE).

The first big difference that the Carrefour Taiwan team noticed after the migration was the faster and more stable connectivity, supported by Google Cloud's data center in Taiwan. When the company's internal systems were deployed in France, the system response was very slow due to the long data transfer distance. Previously, it took over a minute for Carrefour Taiwan's human resource system to complete a leave request procedure. Now, the procedure can be done in less than 10 seconds on Google Cloud. Ting says that the company's work efficiency has been greatly enhanced thanks to the faster system operating speed.

At the same time, the network stability of Carrefour Taiwan's ecommerce platform and mobile app has also improved. Before, the company's two services were hosted in a data center in Hong Kong and frequently encountered network issues such as slow connections or disruptions. As the public cloud platform that Carrefour Taiwan previously used didn't have a local branch office, the retailer was unable to acquire sufficient technical support to identify the root causes of connectivity issues. Since the migration, Carrefour Taiwan's ecommerce platform and mobile app have never experienced

any network problems, delivering a better user experience.

Reducing operational costs while enhancing service availability

When moving its ecommerce platform and mobile app to Google Cloud, the Carrefour Taiwan team changed the structure design of its two services for lower costs and higher availability. Pre-migration, the retailer used virtual machines (VMs) to run the servers and databases of its ecommerce platform and mobile app, which was costly and challenging to manage. By adopting the microservice architecture of Cloud Run and GKE for servers, and Cloud SQL (<https://cloud.google.com/sql>) for databases, Carrefour Taiwan has saved 40% on operational costs, because the new cloud infrastructure is more cost-effective, and Google Cloud offers committed use discounts.

Additionally, the new computing architecture is easier to manage and scale. Prior to the adoption of Google Cloud, Carrefour Taiwan's ecommerce website and mobile app could be down for an entire day due to high traffic, because the company's engineers needed to manually adjust its VMs, and sometimes failed to meet the sudden increase in demand. As GKE and Cloud SQL support autoscaling, and Cloud Run can distribute traffic among instances without disrupting the service, Carrefour Taiwan is able to greatly increase the availability and stability of its services.

"Our ecommerce platform and mobile app used to frequently experience downtimes when we launched promotion campaigns, because our previous infrastructure couldn't swiftly handle sudden flow surges," explains Ting. "With easily scalable products like GKE and Cloud SQL, we can now provide highly stable digital services, and the user reports about system failures have significantly decreased."

Building an integrated data warehouse for broader data analytics

In the latest step of its cloud transformation, Carrefour Taiwan has uploaded all its business data to BigQuery (<https://cloud.google.com/bigquery>) and built an integrated data warehouse. Next, the retailer plans to set up a ML model to predict store stock demand for better retail inventory management. It will also leverage Recommendations AI (<https://cloud.google.com/recommendations>) to send product recommendations that fit its customers' preferences and its product promotion plans.

"Google Cloud has helped us realize many digital transformation goals by generating useful data insights, and providing powerful and reliable cloud infrastructure," says Ting. "With its ever-advancing technologies, we believe that we can unlock more value from our data and continue optimizing our retail services."

"Google Cloud
has helped us
realize many
digital
transformation
goals by
generating
useful data
insights, and
providing
powerful and
reliable cloud
infrastructure.
With its ever-
advancing
technologies, we
believe that we
can unlock more
value from our
data and
continue
optimizing our
retail services."

—Henry Ting, Digital Chief
Technology Officer,
Carrefour Taiwan

